Q1. 12 to 32 blastomeres stage of human embryo is called:

a. Morula
b. Gastrula
c. Blastula
d. Trophoblast
e. Embryoblast

Q2. Which mesoderm is segmented to form somites?

A. Somatopleuric
B. Splanchnopleuric
C. Intermediate cell mass
D. Paraxial
E. cloaca.

Q3. Secondary oocyte completes the second meiotic division:

A. Before birth
B. At birth
C. Before ovulation
D. At ovulation
E. At fertilization

Q4. Which of these is a derivative of ectoderm?

A. Dermis of skin
B. spinal cord
C. Gonads
D. Kidney
E. adrenal cortex

Q5. Which portion of the blastocyst will give rise to the embryo?

A. endoderm
B. cytotrophoblast
C. inner cell mass
D. blastocoel
E. outer cell mass

Q6. Cells that form the three primitive germ layers are derived from which of the following:

A. Epiblast
B. Cytotrophoblast
C. Syncvtotrotrophobist

D. Hypoblast
E. Yolk sac

Q7. Which of the following is not a mesodermal derivative?

A. Bones
B. Kidneys
C. Skeletal muscles
D. Enamel of teeth
E. Dermis of skin

Q8. Which of the following is not an ectodermal derivative?

A. Spinal cord
B. Retina of eye
C. Sensory ganglia of spinal nerves.
D. Pituitary gland
E. Cortex of suprarenal gland

Q9. Which one of the following is derived from Endoderm?

A. Striated muscle
B. stroma of thymus
C. Veins
D. Testes
E. Lymphatics

Q10. Which of the following is not an event in the 3rd week of development?

A. Development of 3 germs layer
B. Development of notochord
C. Development of yolk sac
D. Development of primitive streak
E. Development of tertiary villi

Q11. Nerve of the third pharyngeal arch is?
A. Glossopharyngeal
B. Facial
C. Vagus
D. Hypoglossal
E. Trigeminal

Q12. External acoustic meatus develops from which of the following?
A. 1st pharyngeal arch

B. 1st pharyngeal membrane
C. Tubotympanic recess
D. 1st pharyngeal groove
E. 2nd pharyngeal arch

Q. 13 DNA is duplicated in the cell cycle during the
(A) G2 phase.
(B) G0 phase
(C) M phase.
(D) G, phase.
(E) S phase.

Q. 14 The part of the sperm containing proteolytic enzymes to digest the zona pellucida is the:

A. capacitor
B. acrosome
C. corona
D. head
E. cumulus

Q. 15 During the second week of development, the trophoblast differentiates into:

A. intraembryonic mesoderm
B. ectoderm
C. syncytiotrophoblast
D. yolk sac (secondary)

Q. 16 The first two intraembryonic germ layers to differentiate are the:

A. epiblast and hypoblast
B. ectoderm and hypoblast
C. ectoderm and endoderm
D. ectoderm and mesoderm

Q. 17 The primitive streak first appears at the beginning of the week.

A. first
B. second
C. fourth
D. third
E. fifth

Q. 18 In the third week of human embryonic development:

A. the amnion appears
B. a bilaminar embryonic disc is formed
C. the body stalk moves ventrally and joins with the yolk sac stalk to form the umbilical cord
D. the uteroplacental circulation is established
E. the neural plate is induced by the notochordal process and associated mesoderm

Q. 19 The ovulated mammalian oocyte is arrested at:

A. prophase of meiosis I
B. metaphase of meiosis II
C. prophase of meiosis II
D. metaphase of meiosis I
E. none of the above

Q. 20 Tragus and helix of external ear are derived from:

A. 1st pharyngeal pouch
B. 1st cleft
C. 1st branchial arch
D. 2nd branchial arch
E. 2nd cleft

Q. 21 Failure of the brain to grow may result in:

A. microcephaly
B. craniostenosis
C. acrocephaly
D. scaphocephaly
E. plagiocephaly

Q. 22 As a resident in orthodontics, you are called to see a newborn who has a unilateral cleft lip and a unilateral cleft of the primary palate. This condition is most likely the result of:

A. failure of fusion of the mandibular prominences
B. failure of fusion of the medial nasal processes
C. failure of fusion of the lateral palatine processes with the nasal septum
D. failure of fusion of the maxillary prominence with the medial nasal prominence
E. failure of fusion of the paired lateral palatine processes.

Q.23 The foramen cecum of the adult tongue:

A. marks the point of embryonic evagination of the thymus gland
B. divides the tongue into two parts, an anterior one-third and a posterior two-thirds
C. has no embryological significance

D. develops into taste buds
E. marks the point of embryonic evagination of the thyroid gland

Q.24. Spermatogenesis and oogenesis describe the process of meiosis in males and females, respectively. This statement about both processes is true

(a) both are halted in an intermediate step until puberty
(b) oogenesis produces a larger gamete than spermatogenesis
(c) both produce the same number of viable gametes
(d) oogenesis forms two polar bodies, spermatogenesis forms only one

Q.25 The terminal dilated part of the hindgut is called the:
A. allantois
B. yolk stalk
C. cloaca
D. cecum
E. coelom

Q.26 The prechordal plate will be the future location of:

A. Mouth
B.Stomach
C. Heart
D.Spinal cord

Q.27 The process by which the bilaminar disc is converted into trilaminar disc is:

A.Neurulation
B. Folding
C.Implantation
D. Gastrulation

Q.28 Which of the following is not a true statement with regard to the fetal circulation?

A. Since the fetal liver is a hematopoietic organ, it is large and well supplied with oxygenated blood.
B. Fetal brain receives relatively pure arterial blood.
C. Foramen primum of the interatrial septum closes after the formation of the foramen secundum.
D. In early developmental stages, one pulmonary vein buds from the left atrium of the heart.
E. Fetal and maternal blood vessels anastomose in the placenta

Q.29 The fetal period is:

A.The period from implantation till the 8th week
B. The period from 9th week till birth
C. The period from 2nd-3rd week
D.The period from 7th -11th week

Q.30 The optic nerve is derived from what embryonic tissue?
A. neural crest
B. head mesenchyme
C. ectoderm
D. mesoderm
E. endoderm

Q.31 Branchial cysts or lateral cervical cysts:

A. are found along the anterior border of the sternocleidomastoid muscle
B. are formed from a rupture of the membrane between pharyngeal pouches and branchial clefts
C. are remnants of the thyroglossal duct
D. are found in front of the ear

Q.32. The seven-day blastocyst:

A. has a single layer of trophoblast at the embryonic pole
B. has an amniotic cavity
C. is surrounded by a degenerating zona pellucida
D. is attached to the endometrial epithelium
E. is called the hypoblast

Q.33 During implantation, the blastocyst:

A. implants in the endometrium
B. usually attaches to endometrial epithelium at its embryonic pole
C. usually implants in the posterior wall of the body of the uterus
D. causes change in the endometrial tissues
E. all of the above are correct

Q.34 The early stages of cleavage are characterized by:

A. formation of a hollow ball of cells
B. increase in the number of cells in the zygote
C. increase in the size of the cells in the zygote
D. formation of the zona pellucida

Q.35 Of the following, the item most closely associated with the pons is:
A. telencephalon
B. diencephalon
C. metencephalon
D. mesencephalon
E. myelencephalon

Q.36 During development, the notochordal process:
A. extends from the prochordal plate to the primitive node
B. arises from involuting endodermal cells
C. is involved in the induction of the primitive gut
D. becomes the appendicular skeleton

Q.37 Which of the following is not a component of the mature placental barrier?

A. the endothelial lining of fetal capillaries
B. the basement membrane of fetal capillaries
C. the syncytiotrophoblast
D. the cytotrophoblast
E. all of the above are part of the mature placental barrier

Q.38 Somitomeres, paraxial mesoderm cranial to the somites, do not give rise to which of the following skeletal muscles in the head?

A. extrinsic muscles of the eye
B. masseter
C. muscles of facial expression
D. muscles of mastication
E. temporalis tongue muscles

Q.39 The portion of the decidua which does not survive until the end of pregnancy is the:
A. basalis
B. capsularis
C. laeve
D. parietalis
E. frondosum

Q.40 When chorionic villi become vascularized they are called as:

A. branch villi
B. stem villi
C. tertiary villi
D. anchoring villi

E. mature villi

Q.41 Chorionic villi are designated as secondary chorionic villi when they:

A. develop a mesenchymal core
B. are covered by syncytiotrophoblast
C. contact the decidua basalis
D. give rise to branch villi
E. none of the above

Q.42 The wall of the chorionic sac is composed of:

A. cytotrophoblast and syncytiotrophoblast
B. extraembryonic splanchnic mesoderm and both layers of trophoblast
C. trophoblast and exocoelomic membrane
D. two layers of trophoblast lined by extraembryonic somatic mesoderm
E. none of the above

Q.43 When the amount of amniotic fluid exceeds two liters, the condition is called:
A. oligohydramnios
B. hydrogravida
C. amnionitis
D. bag of waters
E. polyhydramnios or hydramnios

Q.44 Which of the following structures does not turn under onto the ventral surface of the embryo during folding of the head?
A. prechordal plate
B. notochord
C. heart
D. pericardial cavity
E. septum transversum

Q.45 Almost all of the internal organs are well laid down by:
A. 1 month
B. 3 months
C. 2 months
D. 4 months
E. 5 months

Q.46 In the development of the nervous system:
A. the pontine flexure is in a direction opposite the mesencephalic and cervical flexure
B. during the 5th week, each brain vesicle subdivides into 2 parts
C. there are five primary brain vesicles

D. neural epithelial cells are found in the marginal layer

Q.47 Cells from the primitive streak do not become:
A. endoderm
B. intermediate mesoderm
C. paraxial mesoderm
D. amnioblasts
E. lateral plate mesoderm

Q.48 The notochord is replaced by the:
A. ependyma
B. spinal cord
C. spinal canal
D. dorsal roots
E. nucleus pulposus

Q.49 The hypobranchial eminence contributes to formation of:
A. anterior portion of the tongue
B. posterior portion of the tongue
C. musculature of the tongue
D. epiglottis
E. palatine tonsil

Q.50 The fact that general and special sensory information from the posterior part of the tongue is carried by glossopharyngeal nerve indicates that this part of tongue is from branchial arch
A. I
B. II
C. III
D. IV
E. VI

Q.51 Myoblasts from the occipital myotomes are believed to give rise to the muscles of the:
A. tongue
B. face
C. ear
D. jaw, for mastication
E. eye

Q.52 Which of the following are associated with the 2nd pharyngeal arch?

A. the malleus bone
B. the lower portion of the hyoid bone
C. glossopharyngeal muscle

D. facial nerve
E. anterior belly of digastric

Q.53 Which of the following structures is not part of the first branchial arch?
A. malleus
B. mandibular process
C. sphenomandibular ligament
D. maxillary process
E. stylohyoid ligament

Q.54 The brain flexure which develops between the metencephalon and the myelencephalon is called the:
A. hindbrain
B. pontine
C. cervical
D. cephalic
E. midbrain

Q.55 Which structures are derived from the intermaxillary segment of the embryonic face?
A. philtrum
B. anterior portion of the palate
C. anterior portion of the upper jaws
D. upper incisor teeth
E. all of the above are correct

Q.56 Of the following, the one most closely associated with the 3rd ventricle is:
A. telencephalon
B. diencephalon
C. mesencephalon
D. metencephalon
E. myelencephalon

Q.57 Each pharyngeal arch includes:
A. derivatives of ectodermal neural crest cells
B. an aortic arch artery
C. a mesodermal core from paraxial mesoderm
D. a cranial nerve
E. all of the above